

Refrigeration to prevent food losses

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ABSTRACT

Globally, it is estimated that 1.3 billion tonnes of edible food are lost every year, which is one-third of the food available. Food is lost throughout various stages of the food supply chain (production and harvesting, postharvest handling, processing, distribution and consumption), but where the main losses are, differs between countries. In medium- and high-income countries, food is mostly wasted in the kitchen or at the end of the supply chain by consumers. In low-income countries, food is mainly lost during the early and middle stages of the food chain because of the drawbacks in the farming, harvesting, packaging and often a non-existing or poor cold chain. This paper gives data for the food losses in the world, with special focus on China and India. Examples of measures to reduce food losses are given, with focus on improved cold chain.

Keywords: Food loss, Cold chain, India, China

1 INTRODUCTION

Globally, 1.3 billion tonnes of food are lost every year (Gustavsson et al. 2011). Food loss refers to the food resources that go unutilized during the production, handling and processing stages while waste refers to the food thrown away by the consumers. Majority of food waste occurs at the consumption stage in Europe, but in many countries in Asia, most of the loss occurs at the post-harvest stage due to improper handling and cold chain management.

There is an urgent need to provide enough and safe food (not harmful and with all nutrients preserved) for the growing world population, which is expected to increase from 7.6 billion today to 9-11 billion in 2050 (World Population Trends 2017), (FAO 2016). Optimum storage condition for high quality and long shelf life of products in a cold chain depends on temperature and relative humidity of the storage chamber. Fluctuation of these parameters during storage, transportation and shipment is responsible for growth and survival of pathogenic microorganisms. Food is lost or wasted throughout various stages of the food supply chain (shown in Fig. 1):

- i. **Production and harvesting:** Agricultural produce and harvest are damaged or spilled, animals may die or be diseased, milk could be lost due to cattle ailments, and fish may be thrown away.
- ii. **Postproduction handling:** Spoilage during storage, transportation and other postharvest handling activities. Reasons can be high temperature exposure, bacterial infections, mold, pests, insects, etc.
- iii. **Processing:** The quality and shelf life of the food can be lost or degraded during washing, peeling, slicing, canning, packaging etc., or during slaughtering, smoking, freezing or pasteurizing.
- iv. **Distribution:** Wholesale markets, supermarkets, retailers and other stakeholders in food distribution also experience some food loss and waste (e.g., when cans and packages etc. are accidentally opened or destroyed) during the transportation activities.
- v. **Consumption:** The consumers usually waste food by throwing it away. Cooking more food than is necessary, over-shopping for convenience and even overfeeding is listed in this category.

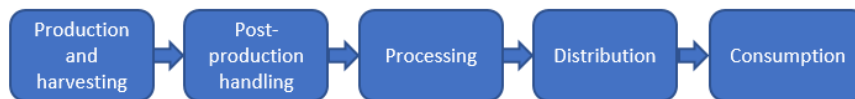


Figure 1. Different stages of food supply chain

2 GLOBAL VIEW ON FOOD LOSS AND WASTE

Several international targets have been set to end hunger and reduce the food wastage. United Nations Sustainable Development Goals (SDG) are probably one of the largest and ambitious targets in human history. The 2nd SDG goal sets its sight towards zero hunger and the 12th goal on responsible consumption and production. The target 2.1 and 2.2 aims at ending hunger and malnutrition by 2030. Similarly, the target 12.3 aims at reducing the food waste and loss by 50 % along the production and supply chains (with post-harvest losses) within 2030. (Lipinski et al, 2017)

2.1 Food loss and waste

According to *Momagri*, China, European Union (EU), United States (USA) and India are the four major food producers in terms of economic values. Fig. 2 illustrates that the total per capita food loss in Europe and North-America is 280-300 kg/year. In sub-Saharan Africa and South/Southeast Asia, it is 120-170 kg/year. The total per capita production of edible parts of food for human consumption is, in Europe and North America, about 900 kg/year and, in sub-Saharan Africa and South/Southeast Asia, 460 kg/year. Similarly, the per capita food wasted by consumers in Europe and North America is 95-115 kg/year, while the figure in sub-Saharan Africa and South/Southeast Asia is only 6-11 kg/year. FAO estimates that 30 % of cereals, 20 % of the dairy, 35 % of fish and seafood, 45 % of fruits and vegetables, 20 % of meat products, 20 % of oil seeds and pulses, and 45 % of roots and tubers are wasted. Looking at the various stages of the food supply chains worldwide, the major producer of food wastage is the phase of agricultural production followed by the postharvest handling and storage and final consumption. (FAO 2012) (Gustavsson et al, 2011)

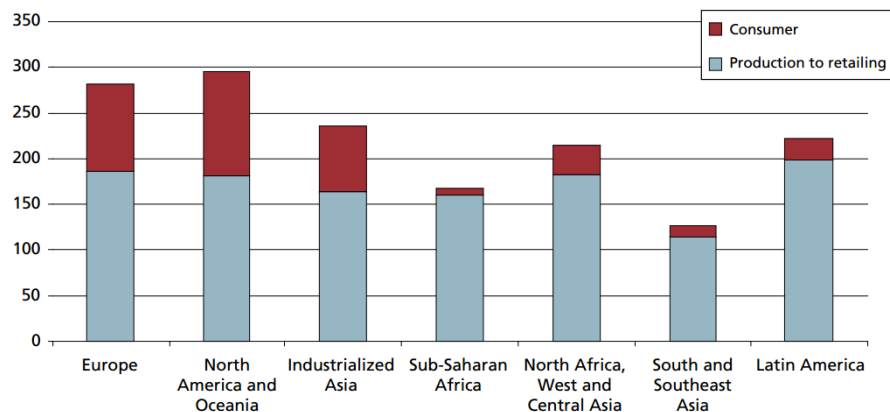


Figure 2. Per capita food losses and waste (Kg/year) (Gustavsson et. al., 2011)

2.2 Case: China

China is the largest producer of food in the world. As of 2010, China produced food worth about € 740 trillion and the food worth € 32 billion was exported, the import was worth € 72 billion. Most of the estimates of the food loss and waste in China are primarily based on the Chinese staple food maize, rice and wheat. Thus, the information on food loss and waste along the various stages of food supply chain is insufficient. Chinese agriculture arrangement is highly decentralized, so a significant percentage of post-harvest losses (5.7-8.6 % grains, 2.5-3.7 % meat, and 10-15 % for perishable food) occur during storage. There is about 6 % food loss during the storage of the food. Even if the governmental storage systems are efficient, it is difficult to use such facilities in a country where farms are dispersed. The on-farm grain storage facilities had losses more than 8 % while the central facilities had losses less than 0.5 % of the grain storage. There are also problems in the controlling of pests, rodents and moulds due to the lack of strong pesticides in the farms and the basic knowledge for storage and technology is lacking for grains storage in many farmers.

In recent years, the industrial development has cut short the harvest and post-harvest losses but the food waste habits in consumers has increased dramatically, due to the increasing affluence in the Chinese society. Consequently € 27-38 billion worth or 17-18 million tonnes of food is wasted annually in China. (Liu, 2013). The Chinese catering and restaurant businesses have been the major source of food waste and not the households. The food loss is as high as 19% in the restaurants, which is surprisingly higher than the 7% of food loss happening in the households. However, only 5% of the food is lost when people eat in the canteens.

2.3 Case: India

India is a major producer and exporter of food products and food processing is recognised as a priority sector in their new manufacturing policy. India is also a major exporter of fruits and vegetables to Europe. Fruits and vegetables losses in India range from 15% for potatoes to up to 50% for citrus fruits (Jakhar, et al. 2015). The total food loss in India amounts to over 26 billion Euros annually, as about 30 percent of all food produced is spoiled after harvesting due to lack of cold chain management and proper food processing units (Dhakal et al 2014). The Indian agricultural sector is highly dispersed compared to other countries. The average farmers have just 2-4 acres of land and 70% of the farmers have less than 1 or 2.5 acres of land. (The CSR Journal, 2015). It is difficult to apply the modern and efficient techniques and technologies in each of these farms because it could be costly. (Artiuch & Kornstein, 2012)

In aquaculture, the total loss in farm operation was about 10%. The losses in the catch of marine fish is because the low value fish is discarded and only the high value fish is taken by the fishermen. However, this is changing now a days as the fishermen bring the low value fish home. The overall total loss of 11 % was observed in which is significantly higher than the previous study (3%) of 2005-07, mainly because the catch operation was not covered in the previous study.

In case of meat, the total loss in farm operations was found to be about 2% due to loss during slaughtering. About 1% of loss happened in the process of storage at the retailers and wholesalers. The overall total loss of meat at national level is about 3%. The poultry meat loss at the national level is about 7% and the losses during the storage of poultry meat is about 4%. It is due to the lack of proper distribution, cold chain system and refrigerated display in the market. Further, the overall total loss of milk in the national level is about 1% only.

3 REFRIGERATION TO PREVENT FOOD LOSSES

Refrigeration includes both chilling and freezing and is one of the best ways of preserving food as close to its original state as possible. When the product temperature is reduced, microbiology activity and chemical and physical changes are slowed down, and shelf life is extended. Improvements to the food cold chain and better use of refrigeration will help to reduce some of the food losses in the world. With a growing population in need of more food, it is necessary to improve the cold chains especially where it is poor today. (IIR, 2009). In the introduction of this paper, the food value chain was divided into 5 stages. In this section, possibilities of improving the cold chain for each stage will be given.

3.1 Production and harvesting

During production and harvesting, there is seldom a need for refrigeration, but it is important that some highly perishable food stuffs, for example fish and seafood, are quickly chilled or frozen after production and harvesting. If not, the quality degradation starts, and it is not possible to go back to the original state. Handling of food after production and harvesting is described in the following section.

3.2 Postproduction handling

Many **fruits, vegetables, tubers and roots** must be stored above freezing temperatures to avoid damage. Examples are bananas (13-15 °C), lemons (10-13 °C), cucumber (10-12 °C), mangos (13 °C), tomatoes (8-13 °C), sweet potatoes (13-15 °C) and potatoes (4-15 °C) (ASHRAE, 2006, chap. 11). However, chilling of these products is still important to prevent them from becoming overripe, especially at high ambient temperatures. A refrigeration system will mainly control the temperature,

but also humidity when necessary. When transporting unwrapped products, more focus should be on humidity control because unwrapped products will dry faster than products that are packed.

Cold water **fish and seafood** will degrade fast if not chilled quickly after harvesting. Fish and seafood from warmer waters will not have the same degradation rate but must also be chilled to preserve quality and shelf life. An example is the development within the pelagic fish industry in Norway, which increased drastically during the 1960s. The fishing vessels became larger and more effective in catching fish. The fishing grounds were also farther away from the shore. However, the traditional way of storing the fish in boxes with ice turned out to be too labour-intensive which often led to increased fish temperatures during transportation. This resulted in poor quality and a lot of the catch had to be sold as oil or fishmeal, with very low profit. After some time with searching and testing, fish tanks with refrigerated seawater (RSW) were selected as the best method for preserving the products on the fishing vessels. In Norway, the primary requirements included insulated tanks, chilling to $-1\text{ }^{\circ}\text{C}$, circulation, cleaning equipment, loading equipment and temperature controlling equipment. The onboard RSW chilling resulted in that almost all mackerel and herring were sold to consumers and not being processed to fish oil or meal. (Widell et al, 2016). Other types of fish, like cod and saithe, are typically stored fresh on ice in the ship. In most cases, there is not a refrigeration system onboard, so the ice is brought from land. Enough ice should be brought to last for the complete journey. These types of fish can also be frozen onboard, typically in plate freezers or filleted and single-frozen.

Animals for **meat** production are mainly transported alive to the slaughterhouse. Chilling of the meat after slaughter and partitioning is necessary, but the rate of chilling affects the quality of the product. Chilling too rapidly will decrease the tenderness of the meat (cold-shortening), especially for beef, pork and lamb. However, electrical stimulation of the meat can be used for avoiding this issue. Studies show that the weight loss can be reduced by fast chilling of meat. (Johannessen, Paulsrud, 2000)

Milk cooling and storage at farms has been provided by innovative R744 refrigeration units in Norway and elsewhere (Fig. 3, left). Such units provide a fast cooling of fresh milk, while the hot side of the refrigeration circuit is applied to produce sanitary hot water at 90°C , applied for cleaning purposes of the milk handling equipment.



Figure 3. Milk cooling unit with CO₂ refrigeration system (left). Mackerel fillets frozen in belt freezer (right)

3.3 Processing

Many different food products are further handled in an industrial processing plant, before being distributed to the consumers. Typical processing includes peeling, cutting and chilling/freezing of **fruits, vegetables, tubers and roots** and slaughtering, gutting, partitioning, filleting of **fish and meat**. Refrigeration during all steps of processing will increase quality and could also reduce food loss.

Superchilling is a process where the temperature of a product is reduced below regular chilling, but still higher than freezing store temperatures. The product is partly frozen, but the consumer will experience it as a fresh product. Chilling a product below initial freezing point will result in a frozen surface layer of 1-3 mm (Kaale, 2014), which creates a cold storage within the product. This can be compared with chilling with ice, where the cold storage is outside of the product. The shelf life of the product increases with decreasing temperature, especially at lower temperatures. During transport, the temperature of the product will level out, resulting in a low temperature within the product and

less or no ice. In general, fast cooling results in smaller and more even distribution of the ice crystals than slow cooling, which will give a better product quality. Fast cooling can also give higher production capacity. Other factors that affect the quality of the products are the initial quality of the product, packing and storage temperature. The complete cold chain should therefore be evaluated, to get the best possible product.

Superchilling can be a less energy demanding alternative to a freezing/thawing, with less liquid loss. It also includes fewer handling elements and thus lower labour costs (Magnussen, Haugland et al. 2008). When storing and transporting chilled products, it is common to do this with ice together with the fish inside the boxes. A typical ratio is 1/3 of ice and 2/3 of fish. When superchilling the fish, ice is not necessary, since parts of the products are frozen. That means more product within each transport, which is beneficial both financially and environmentally. The challenge with superchilling is to control the quantity of ice in the product, the ice fraction, so that it is neither too low, nor too high.

Frozen food products are mainly stored between -20 °C and -30 °C. The lower temperature levels are necessary for fish and ice cream. There are several different food freezing methods. Air is often used because it is easily available, even though it has poor heat transfer properties. Air blast fans increase heat transfer, but they also use electricity and add heat to the freezing tunnel. Large products are often frozen in air blast freezing tunnels, while smaller products are frozen in belt freezers (see Fig. 3, right) or fluidized bed freezers. If the product is unwrapped, a disadvantage is evaporation from the surface. Packaging, on the other hand, results in poorer heat transfer and longer freezing times. There is a risk of uneven air distribution and uneven freezing times in a blast freezer, which can partly be avoided if the products are continuously loaded and unloaded. Proper tunnel design and loading is essential. Unpacked, single products can be frozen in impingement freezers at high production rates. In these, air jets (20-40 m/s) are directed at the products, which provides higher heat transfer than in blast freezers. (Kaale et al., 2014) (Widell, 2012)

Another type of freezing system is the plate freezer, where the refrigerant is circulated inside plates and the products are pressed between the plates. The products are frozen into equally shaped boxes, which is beneficial for stacking and transport. These freezers are energy efficient because no fan energy is needed, and higher evaporation temperatures can be used than in air blast freezers. Liquid as a cooling media is used in cryogenic freezers. The products are sprayed with, for example, liquid carbon dioxide resulting in high temperature differences and high production rates. A refrigeration plant is not needed, but the cooling media is expensive, and large quantities are necessary. All types of freezers should be insulated from surrounding (warmer) areas to prevent heat leakage into the freezer.

Heat absorbed by the refrigeration unit during the cooling / freezing process can be upgraded to e.g. useful hot water/steam applied to clean and sterilize the process equipment after usage. Depending on the simultaneity of cooling and heating demands, compact thermal cold/hot storage devices can be integrated into the energy systems.

3.4 Distribution

To secure food quality and at the same time achieve a high seasonal energy efficiency of the refrigeration systems, the performance requirements should be described over a wide range of operating conditions. High part load efficiency is important for both stationary equipment as well as for transport refrigeration units. Implementation of capacity control devices and thermal energy storage (for example phase change materials) to avoid on-off operation are examples to secure constant foodstuff temperatures. Reliability and redundancy options are important factors to secure the process plant operation.

Transport of food between the production sites, after processing and storage towards the end-users must be performed in well designed and refrigerated containments. Temperature fluctuations should be avoided.

The supermarket or local grocery store is the final distribution centre for food towards the end-users. To avoid food wastage of chilled and frozen foodstuff at this stage of the food chain, the temperature distribution and variations within the cold rooms and display cabinets are the essential parameters (Minetto et al. 2014). Novel CO₂ refrigeration units enable uniform temperature distribution and

eliminate the need for long defrosting cycles. Long cycles of defrosting could lead to several degrees of increased product temperatures, reducing product quality and shelf life.

3.5 Consumption

To achieve SDG, appropriate food waste mapping techniques and technologies are being developed and the proper reporting and the compilation of the statistics has been carried out. The 'ForMat' project in Norway carried out 2010-2015 shows a decrease of food waste by 25500 tonnes. Consumers were for example instructed to have the right temperatures in refrigerators, information was given about food waste and that food can be edible also after the "best before"-date. After the focus on consumers in the *ForMat* project, the focus has now shifted towards processing phase, for example a SINTEF Ocean is leading a project on mapping of food loss in seafood industry.

In China, some steps have been started by the central and local government to decrease and manage food wastage. For example, "Empty your plate" campaign in 2013 for highly grassroots level has encouraged the consumers to avoid food waste. A blog of posts in Weibo (a Chinese social media) has enhanced awareness all over the country and changes in the policies so that the restaurants had to change the size of the plates, etc. (Liwei et al., 2013)

Some creative ideas have been implemented in India. For example, the employees of municipality of Kolkata collect the waste directly from the homes and recover 80% of the waste. Women's associations collect and dry the food waste for recycling with minimal charge in cities of Bangalore and Pune. The 419 slums of Mumbai have adopted a system of waste collection effectively. The leftovers from the hotels are distributed to the homeless and orphans. The food not suitable for human consumption is given to the pig farms, the pig farmers collect the remains from the hotels and restaurants.

Many small vendors in India take fruits and vegetables to the doors and sell them. In this process the food is exposed to high temperature, which is the cause of short life of the food products. To cool the vegetables, they sprinkle cold water, but it also leads to faster spoilage of the vegetables. Indian Agricultural Research Institute (IRAI), New Delhi has developed a solar powered vending cart-technology to boost the economy of poor vegetable vendors. It is expected to store the fresh fruits and vegetables for 2-5 days. The extended shelf life of the food can hence help the vendors generate more income. There is a large potential for further development in this area.

3.6 Monitoring environmental parameters

Monitoring environmental parameters such as temperature and humidity in the supply chain of perishable foods is essential to track quality deterioration and thus reduce losses. Real-time monitoring of food quality has a lot of prospect, but it is yet to come to actual practice. However, recent initiatives in this direction indicate the possibilities of real time monitoring and building models to predict and improve quality and shelf life. (Abad et al 2009). Initial conditions of the product also play a key role to assure durability. Often, fruits and vegetables are bruised and damaged due to improper handling during harvesting and transportation (FAO 2017). In this regard, visible and near-infrared (Vis/NIR) spectroscopy has been one of the most successful techniques for non-destructive measurement of chemical constituents and quality attributes (such as damage, internal quality, total soluble solids, titratable acidity and maturity index) of horticultural and food products (Pu et al 2015) (Hu et al 2017).

More attractive solutions than conventional RFID have been proposed in some pilot business research. An example is wireless sensor network (WSN), which can be used to detect different kinds of environment parameters, such as the temperature, humidity, carbon dioxide, ethylene, and vibration (Wang et al 2015). This can help real time tracking and remote monitoring in food traceability. Many authors have proposed optimization models for maximizing the profit in such supply chains, while maintaining quality and reducing losses.

Other factors that can be studied are the degree of impact the products have on the climate and the environment. Winther et al (2009) showed that seafood products exported as frozen or superchilled (shipped) had lower environmental impact than fresh, transported with air freight. Increasing the amount of processed fish, leaving the rest raw material for other production, would also reduce the environmental impact, compared to transporting the whole fish.

3.7 Refrigerants

Refrigeration systems should only apply working fluids with lowest possible climate and environmental impact, therefore, natural working fluids are a safe and sustainable option, due to their well-known behaviour when released into the environment. When developing new heat pumping system, focus should be: energy efficiency also during part load and integration of multi-functionalities. This means that advanced integrated units can provide freezing, cooling, air-conditioning, space heating, hot water, and process steam.

Due to the Kigali Amendment of the Montreal Protocol, and the EU F-gas regulation, the traditional HCFC and HFC refrigeration equipment in the supermarket sector is nowadays replaced by the end-users with units applying natural working fluids. The environmental impact of working fluids like propane (self-contained cabinets) and carbon dioxide (centralised units) are well known and no restrictions will appear in the future. This makes this technology attractive to end-users whenever they can find a local vendor to install and maintain the units. A UNIDO supported demonstration supermarket in the capital of Jordan (Jordan is defined as a High Ambient Temperature HAT country) is an example, that a leapfrog in refrigeration technology from HCFC to CO₂ is possible. The feedback from the supermarket owner after the first year of operation (January 2018 to January 2019):

- No extraordinary maintenance visits, i.e. no failure in the system nor the cabinets
- Zero food wastage due to stable temperature levels in the display cabinets
- Reduction of energy demand related to the refrigeration unit by more than 30% compared to similar new HFC installations in the region

Training and knowledge transfer are important for a successful and fast phase in of natural refrigerants within the food sector and supermarkets globally. Relatively high first costs to install integrated CO₂ refrigeration systems are still the main barrier to a fast phase in of this technology on a global scale, especially when none of the local vendors has delivered such units before.

4 CONCLUSION

With the world population set to approach an estimated 9 billion by 2050, against a background of finite natural resources, we need renewable biological resources for securing healthy food and animal feed to meet the nutritional requirements of the growing population.

Most of high-income areas have good infrastructure and technologies for cold chain management. It is in low-income countries and especially within agriculture that substantial improvements can be made to the cold chains to reduce food loss as a result.

If food wastage was a country, it would be the third largest GHG emitter in the World after China and the USA. Environmental impact of the food chains will be reduced if the overall losses are reduced. Food production requires energy, so reducing food losses will also result in reduction in energy use per tonne of food produced. Energy use has different environmental impacts depending on how it is produced and transferred, but it is seldom zero and improving the energy efficiency of a process is always positive for the environment. Refrigerant systems still use working fluids that, if released to the atmosphere, deplete the ozone layer and increase the global warming potential. During the last decades, there has been a focus on reducing these emissions and finding more environmentally friendly replacements. However, there are still too many refrigeration systems in place that use harmful refrigerants. In further research and development, it is necessary to:

- Reduce food losses along food value chains by:
 - o Increasing awareness of the benefits of high-performance cold chains
 - o Increasing collaboration between politicians, refrigeration stakeholders and food producers to improve cold chain technology and management
 - o Transfer of high-performance, environmentally friendly and cost-effective technology to areas where food loss is high
 - o Training of technicians and users for proper and efficient operation of refrigeration systems
- Reduce energy consumption and carbon footprint in the cold chain

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